

model, but they do not prove a regularisation mechanism or statistical superiority across training initialisations. For SOH, the distribution is broader, which is consistent with the greater difficulty of estimating capacity fade from operational data. Nevertheless, most errors remain within an approximately $\pm 2\%$ band around the reference trajectory.

Global metrics alone would still hide local failures. The chapter therefore also zooms into a high-dynamic pulse segment between 30k and 40k seconds (Figure 7.10) and into a representative check-up sequence between 657k and 897k seconds (Figure 7.11). In the pulse region, the quantized model follows the reference transients with high fidelity and captures voltage-recovery phases accurately, whereas the pruned model smooths some sharper local features. In the check-up phase, all variants remain stable over the extended charge and discharge interval. These local views complement the segment-wise temporal analysis rather than replacing it.

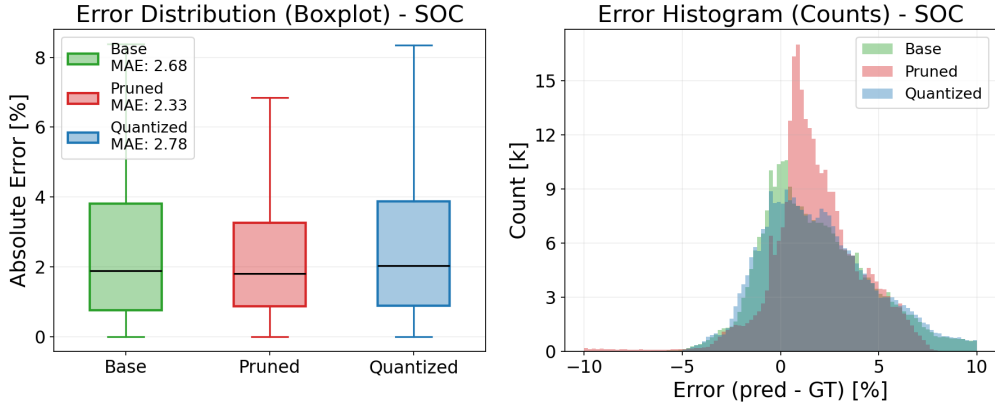


Figure 7.8: SOC error distribution for the embedded SOC deployments. The absolute-error boxplots on the left show that the Pruned model has the smallest median and interquartile range. The signed-error histograms on the right show the distribution relative to the SOC reference and the substantial overlap between the three variants.